

Welcome to Verilog RTL

In digital design you will write register-transfer level code

Why RTL matters

- Coursework is not only drawing block diagrams
- You need modules with ports, wires and registers
- Later labs, counters, shift registers, lint
- Getting comfortable here pays off in every later design and verification course

Two tracks, one idea

- Every lab module offers two ways to practice
- Style rules feel like work you will keep
- The browser lab track uses interactive tools on the learning platform
- You may do either track, or both
- The usual rhythm is browser first for the idea, then a tiny dot-v sketch for muscle memory

Set up the real Verilog track

- Open a text editor for dot-v files, VS Code, Cursor, or vim all work
- Clone or open this course repo and skim the examples folder when a module points you there
- Optionally confirm a simulator is reachable
- You do not need a full EDA install on day one

Set up the browser lab track

Digital Design and Verification Platform

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Courses, labs, tools...

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TOOLS

Interactive labs

Concept labs in the browser — organized by idea, not by course or module. Shipped tools are ready to use; planned tools show **COMING SOON** until they ship. Catalog: **202 shipped, 0 planned** (see [tools.md](#)). Target courses: [syllabus.md](#).

Shell & filesystem

Virtual filesystem terminal

pwd, ls, mkdir, cp/mv/rm · 22 challenges

Permissions, umask, PATH & ownership

rw·x · umask · PATH · 22 challenges

Shell history & reverse-search

history · Ctrl+R · !! / !n / !\$ · 22 challenges

Absolute vs relative paths

Tools index

How to move through modules

- For each module
- SystemVerilog design constructs and UVM are later courses
- When you finish this intro checklist, continue to the first lab module